Unless explicitly requested by a problem, do not include sketches as part of your proof. You are free to use the result from any problem on this (or previous) assignment as a part of your solution to a different problem even if you have not solved the former problem.

Problem 1 (2 pts)**.** Fix an integer $n \geqslant 2$. Prove that a sequence of integers $d_1, \dots, d_n$ is the degree sequence of some tree if and only if $d_i \geqslant 1$ for all $i \in \{1, \dots, n\}$ and $\sum_{i=1}^{n} d_i = 2n - 2$.

Solution. ($\Rightarrow$) Assume $d_1, \dots, d_n$ is the degree sequence of a tree T. Since T is connected, there are no isolated vertices, thus $d_i \geqslant 1$ for all $i \in \{1, \dots, n\}$. Also, by the handshaking lemma, we have

$$\sum_{i=1}^{n} d_i = \sum_{v \in V(T)} \deg v = 2|E(T)| = 2(n-1) = 2n - 2.$$

($\Leftarrow$) We proceed by induction on n. The base case $n = 2$ we have $d_1 + d_2$ with $d_1, d_2 \geqslant 1$, thus $d_1 = d_2 = 1$ and the sequence $(1, 1)$ is the degree sequence of the clique K_2, which is a tree.

Suppose now $n \geqslant 3$. Without loss of generality, we assume that $d_1 \geqslant d_2 \geqslant \cdots \geqslant d_n$.

Claim: $d_n = 1$ and $d_1 \geqslant 2$. Indeed, we have

$$nd_n \leqslant \sum_{i=1}^{n} d_i \leqslant nd_1 \quad \Rightarrow \quad nd_n \leqslant 2n - 2 \leqslant nd_1 \quad \Rightarrow \quad d_n \leqslant 2 - \frac{2}{n} \leqslant d_1.$$

Using $n \geqslant 3$ we get $d_n \leqslant 2 - \frac{2}{n} < 2$; since $d_n \geqslant 1$ we get $d_n = 1$. Also $1 < 2 - \frac{2}{n} \leqslant d_1$ implies $d_1 > 1$, thus $d_1 \geqslant 2$. This proves the claim.

Consider now the sequence $f_1, \dots, f_{n-1}$ with $f_1 = d_1 - 1$ and $f_i = d_i$ for all $i \in \{2, \dots, n-1\}$. Notice that $f_i \geqslant 1$ for all $i \in \{1, \dots, n-1\}$ and also

$$\sum_{i=1}^{n-1} f_i = (d_1 - 1) + \sum_{i=2}^{n-1} d_i = \sum_{i=1}^{n-1} d_i - 1 = \sum_{i=1}^{n} d_i - 2 = 2n - 2 - 2 = 2(n-1) - 2.$$

Thus, by inductive hypothesis, there is a tree T' with $V(T') = \{v_1, \dots, v_{n-1}\}$ and $\deg_{T'} v_i = f_i$. Let T denote the graph obtained from T' by adding a new vertex v_n and a new edge $v_1 v_n$. Note that T is connected, because T' is connected. Also T contains no cycles, because we are adding a new leaf to T'. Thus T is a tree. Moreover, for $i \in \{2, \dots, n-1\}$ we have $\deg_T v_i = \deg_{T'} v_i = f_i = d_i$, for $i = 1$ we have $\deg_T v_1 = \deg_{T'} v_1 + 1 = f_1 + 1 = d_1$, and for $i = n$ we have $\deg_T v_n = 1$. Thus T is a tree that realizes the sequence $d_1, \dots, d_n$. □

Problem 2 (2 pts)**.** This problem gives an alternative proof for the existence of spanning trees.

Let G be a connected graph. Consider the family $\mathcal{F}$ of all spanning subgraphs of G that contain no cycle. Show that $\mathcal{F}$ is non-empty and that an element of $\mathcal{F}$ with **maximum** number of edges is a spanning tree of G.

Solution. Note that $\mathcal{F}$ is non-empty because it contains the empty spanning subgraph $(V(G), \varnothing)$. Let T denote an element of $\mathcal{F}$ with **maximum** number of edges. We show that T is a spanning tree.

Firstly, since $T \in \mathcal{F}$, we know that T is a spanning subgraph of G which contains no cycles; thus we need only show that T is connected. Suppose for the sake of contradiction that T is disconnected; thus we can partition $V(T) = A \sqcup B$ with A and B non-empty such that there are no edges of T between A and B. Since $V(T) = V(G)$ and G is connected, there must be some edge $e \in E(G) - E(T)$ such that e has one end-point in A and the other in B; consider $T' = T + e$. Firstly, T' is still a spanning subgraph of G since we did not modify the vertex set. Next, T' is still acyclic since the edge e must have had its end points in different connected components of T. But this means that $T' \in \mathcal{F}$, a contradiction since T' has strictly more edges than T does. □

Problem 3 (2 pts)**.** Let G be a connected graph and let S be any subset of edges of G. Show that G has a connected, spanning subgraph H such that $S \subseteq E(H)$ and if C is a cycle in H, then $E(C) \subseteq S$.

Solution. Consider the family $\mathcal{F}$ of all spanning subgraphs H of G such that H is connected and $S \subseteq E(H)$. We note that $\mathcal{F}$ is non-empty, since it contains G. Thus, let $H \in \mathcal{F}$ be any element with the fewest number of edges. We claim that H is our desired subgraph.

Firstly, since $H \in \mathcal{F}$, we know that H is a spanning subgraph of G, is connected and $S \subseteq E(H)$. So we need only show that H does not have any cycles using edges not in S. Suppose for a contradiction that C is a cycle in H with $E(C) \nsubseteq S$. Therefore, there is some edge $e \in E(C) - S$; consider $H' = H - e$. Since we did not modify the vertex set by removing e, H' is a spanning subgraph of G. Additionally, $S \subseteq E(H')$, since $e \notin S$. Finally, H' is still connected since e was chosen to be in a cycle of H. But this means that $H' \in \mathcal{F}$; a contradiction since H' has strictly fewer edges than does H. □

Problem 4 (2 pts)**.** Let G be a connected graph and let $w\colon E(G) \to \mathbb{R}^{>0}$ be a weight function. Show that if all weights are distinct (that is $w(e) \neq w(s)$ for all distinct $e, s \in E(G)$), then G has a *unique* minimum spanning tree.

Solution. Since G is connected, we know that G has at least one minimum spanning tree. Thus suppose for a contradiction that there are two distinct minimum spanning T_1 and T_2. Since T_1 and T_2 are different, we know that $E(T_1) \neq E(T_2)$. Thus, the *symmetric difference*

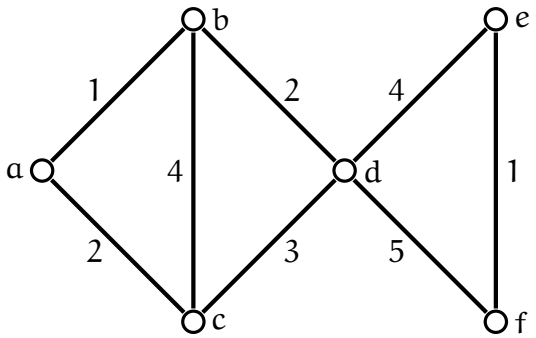
$$E(T_1) \triangle E(T_2) = \big(E(T_1) \backslash E(T_2)\big) \cup \big(E(T_2) \backslash E(T_1)\big)$$

is non-empty. Let $e \in E(T_1) \triangle E(T_2)$ be the smallest weight edge in this set. Without loss of generality, we may suppose that $e \in E(T_1) \backslash E(T_2)$.

Consider the graph $H = T_2 + e$. Since $e \notin E(T_2)$ and T_2 is a tree, we see that H must contain a cycle C which uses the edge e. Now, since T_1 contains no cycles, there must be some edge $s \in E(C) \backslash E(T_1)$; note however that $s \in E(T_2)$. Set $T_3 = H - s = T_2 + e - s$, which is a spanning tree since we removed an edge from the cycle C in H.

Now, $s \in E(T_2) \backslash E(T_1) \subset E(T_1) \triangle E(T_2)$ and so $w(s) > w(e)$ since e was chosen to be the smallest weight element of this set and all weights are distinct. But then T_3 is a spanning tree with $w(T_3) = w(T_2) + w(e) - w(s) < w(T_2)$ which contradicts the fact that T_2 is a minimum weight spanning tree. □

Problem 5 (1+ 1 pts)**.** Consider the following weighted graph

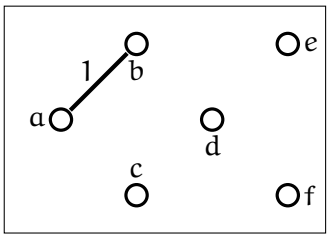


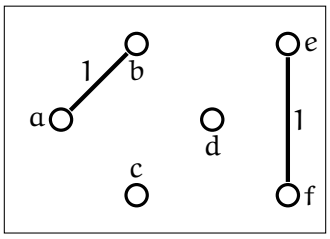
(a) Apply Kruskal's algorithm to find a minimum spanning tree of the graph.

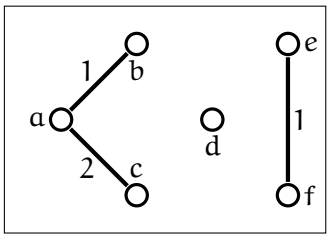
(b) Apply the Jarník-Prim algorithm to find minimum spanning tree of the graph.

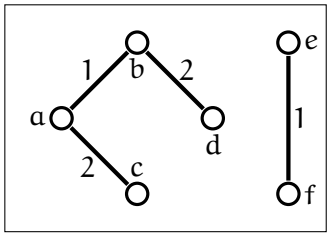
For each case, include a series of pictures explaining how the minimum spanning tree is created.

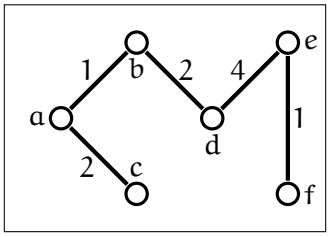
Solution. (a) There solution is not unique, it depends on how we order edges of the same weight. Here, we chose the order: ab, ef, ac, bd, cd, bc, de,df.



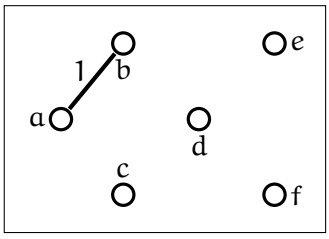


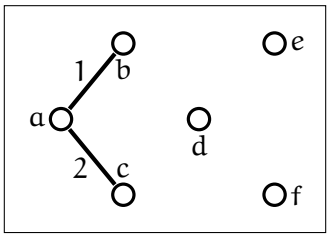


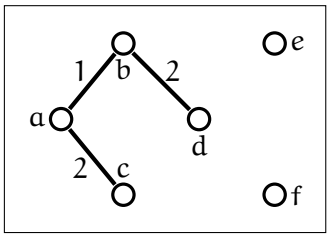


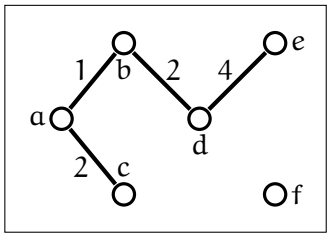


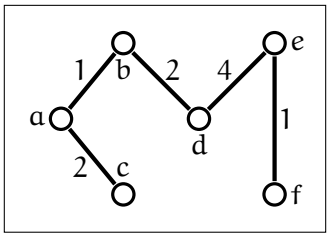
(b) Again, the result will vary depending on which vertex is chosen as the starting tree. Here we choose a as starting vertex..











□